SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5101
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I N. Akshaya (192425129) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: N. Akshaya

95/100

Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis
Given:
p = 17, q = 19

Compute n and $\phi(n)$
Choose e = 5 and calculate the private key d
Encrypt the message M = 7
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

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2. ElGamal Encryption Analysis
Given:
p = 23, g = 5, private key x = 7, random k = 3, message M = 10

Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

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3. RSA Digital Signature Verification
Given:
p = 11, q = 17, e = 7, message M = 8

Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

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4. Diffie-Hellman with Attack Resistance Evaluation
Given:
p = 29, g = 2
User A private key = 5
User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% 20	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% 20	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% 20	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 15	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

95

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CO2 - AT1 - Analytical Problem Solving

1) RSA Key Generation and Security Analysis

Given

*P = 17, q = 19

* public exponent e = 5

* Message M = 7

Step 1: Calculate n and $\phi(n)$

RSA modulus:

$n = p \times q$

$n = 17 \times 19$

$n = 323$

Euler's toient function:

$\phi(n) = (p-1)(q-1)$

$\phi(n) = 16 \times 18$

$\phi(n) = 288$

Step 2: Calculate the private key d

The private key satisfies:

$d \times e \equiv 1 \pmod{\phi(n)}$

$5d \equiv 1 \pmod{288}$

using the extended Euclidean algorithm:

$1 = 5(173) - 3(288)$

$\therefore d = 173$

Public key $(e, n) = (5, 233)$

Private key $(d, n) = (173, 323)$

Step 3: Encrypt the message

RSA encryption formula:

$C = M^e \bmod n$

$C = 7^5 \bmod 323$

since $7^5 = 16807$

$16807 \bmod 323 = 17$

$\therefore$ The ciphertext is $C = 17$

Step 4: Decrypt the ciphertext

RSA decryption formula:

$M = C^d \bmod n$

$$M = 17^{173} \bmod 323$$

using modular exponentiation,

$$M = 7$$

The decrypted message matches the original message.

Step 5: Security Analysis

Small prime numbers produce a small RSA modulus, which is easier for an attacker to factor. If the attacker to factor. If the attacker finds p and q, the attacker can calculate $\phi(n)$ and derive the private key.

∴ Real RSA systems use much larger keys and secure padding schemes.

Result: RSA encryption and decryption were successfully performed, and the original message 7 was recovered.

2) ElGamal Encryption Analysis

Given

* prime number $p = 23$

* Generator $g = 5$

* private $x = 7$

* Random value $k = 3$

* Message $M = 10$

Step 1: Calculate the public key

The public key component is:

$$y = g^x \bmod p$$

$$y = 5^7 \bmod 23$$

using modular exponentiation:

$$5^2 \bmod 23 = 2$$

$$5^4 \bmod 23 = 4$$

$$5^7 \bmod 23 = (5^4)(5^2)(5) \bmod 23$$

$$y = (4 \times 2 \times 5) \bmod 23 = 17$$

Public key: $(p, g, y) = (23, 5, 17)$

Private key: $x = 7$

Step 2: Encrypt the message

ElGamal generates two ciphertext components.

First component:

$$C_1 = g^k \bmod p$$

$$C_1 = 5^3 \bmod 23 = 10$$

Second Component:

$$C_2 = M \times y^k \bmod p$$

$$C_2 = 10 \times 17^3 \bmod 23$$

Since

$$17^3 \bmod 23 = 14$$

$$C_2 = (10 \times 14) \bmod 23 = 2$$

∴ The ciphertext pair is:

$$(C_1, C_2) = (10, 2)$$

Step 3: Decrypt the ciphertext

Calculate the shared value:

$$S = C_1^{x} \bmod P$$

$$S = 10^{7} \bmod 23$$

$$S = 14$$

Find the modular inverse of modulo 23

$$14^{-1} \bmod 23 = 5$$

This is because:

$$14 \times 5 = 70 \equiv 1 \pmod{23}$$

Now decrypt:

$$M = C_2 \times S^{-1} \bmod p$$

$$M = 2 \times 5 \bmod 23 = 10$$

The original message analysis is recovered Correctly.

Step 4: Security analysis of randomnes

The random value k helps produce different

Ciphertexts when the same message. The random value $k = 3$ plays an important role in ElGamal encryption. using different random values for the same message produces different ciphertests, making it difficult for attackers to identify repeated messages. The value k must be generated securely and unpredictably.

Reusing the same k can weaken security and expose secret information.

Result:

The ciphertest $(10, 2)$ was decrypted success. fully to cover the original message 10.